



[MCT312s]

Industrial Automation

Team 1

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Table of Contents

Table of Figures	2
Responsibility sheet and task allocation for each team member	3
Project Overview:-	4
The sequence of our project:	5
The production line In Factory I/O :.....	6
The production line components:	7
1- feeding system:.....	7
2- Sorting station 1.....	8
3- Machining Center station	9
4- Sorting Station 2	10
5- Control Panels	10
6- Assembly Station	11
HMI (Human Machine Interference):	12
Conclusion :-	13
Drive Links:-	14

Table of Figures

Figure 1 Production Line	6
Figure 2 Feeding System.....	7
Figure 3 Sorting Station 1	8
Figure 4 Machining Center station	9
Figure 5 Sorting Station 2	10
Figure 6 3 Control Panels For 2 Assembler and the Production Line	10
Figure 7 Assembly Station	11
Figure 8 HMI Screen	12
Figure 9 States HMI	13

Responsibility sheet and task allocation for each team member

NO	Name	Percentage %
1	Abdelkarim Wael Abdelkarim	20
2	Mahmoud Salama Mohamed Abdo	20
3	Musa Gamal El-Shenawy	20
4	Abdullah Mahmoud Abdelkarim	30
5	Youssef Ayman	5
6	Sam Isaac	5

Project Overview:-

The goal of this project is to design and simulate a fully functioning production line on a simulation platform, such as FACTORY I/O, that closely mimics real industrial processes. The project will incorporate considerations for the various stages of the production process, including sorting, machining, and assembly, as well as any other relevant factors.

To control the simulated production line, the project will use Siemens TIA portal software, a powerful tool for programming and configuring automation systems. The software will be used to develop a customized automation system that is tailored to the specific needs of the simulated production line, with an emphasis on reliability, efficiency, and ease of maintenance.

To provide a visual representation of the various states of the simulated production line, the project will also include the design of a suitable HMI (human-machine interface) module. This will allow users to easily monitor and control the production line, as well as identify any potential issues or bottlenecks in the system.

The sequence of our project:

- 1)The production line starts when we press start button.
- 2)Our production line starts with an emitter which emit random color (green or blue) Raw material.
- 3)the raw material moves on the conveyors and when the diffuse sensor detect it the pick and place robot starts its work then the material enter the sorting station 1.
- 4) the sorting station 1:-
 - in this station we use vision sensor to detect the colour of the raw material so we can sort the Blue in the left path by a pusher and the Green continue in the right path then they enter to the Machining center station(we have two CNC one for green[CNC1] and one for blue[CNC]) the first product will be base and the next raw material enter the CNC will become lid and so on then we have sorting station.
- 5) the sorting station 2:-
 - in it we use also a vision sensor to detect if the product is lid or base to sort the base in the right base and wait in the assembly station for the lid as it clamps and the lid will go on the left path when it reach the assembly station(we have two assembly stations (one for green and one for blue) the two axis pick and place starts its work and assembly the lid and base The final product will continue to the finish conveyor then calculate the number of Final Product .

The production line In Factory I/O :

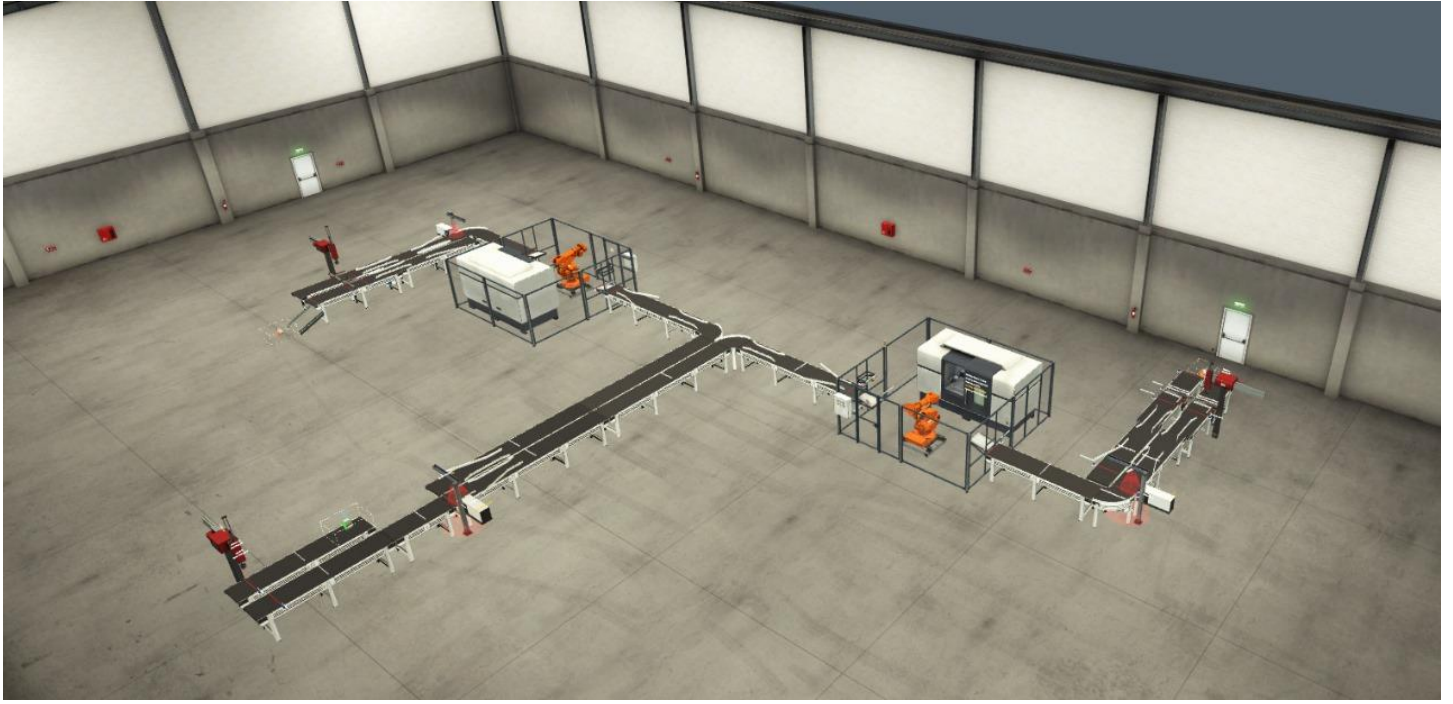


Figure 1 Production Line

The production line components:

1- feeding system:

- This Unit is the beginning of our production line, where it generates random raw materials. It also incorporates a pick & place robot which can be used to place the fed part at the beginning of your line.

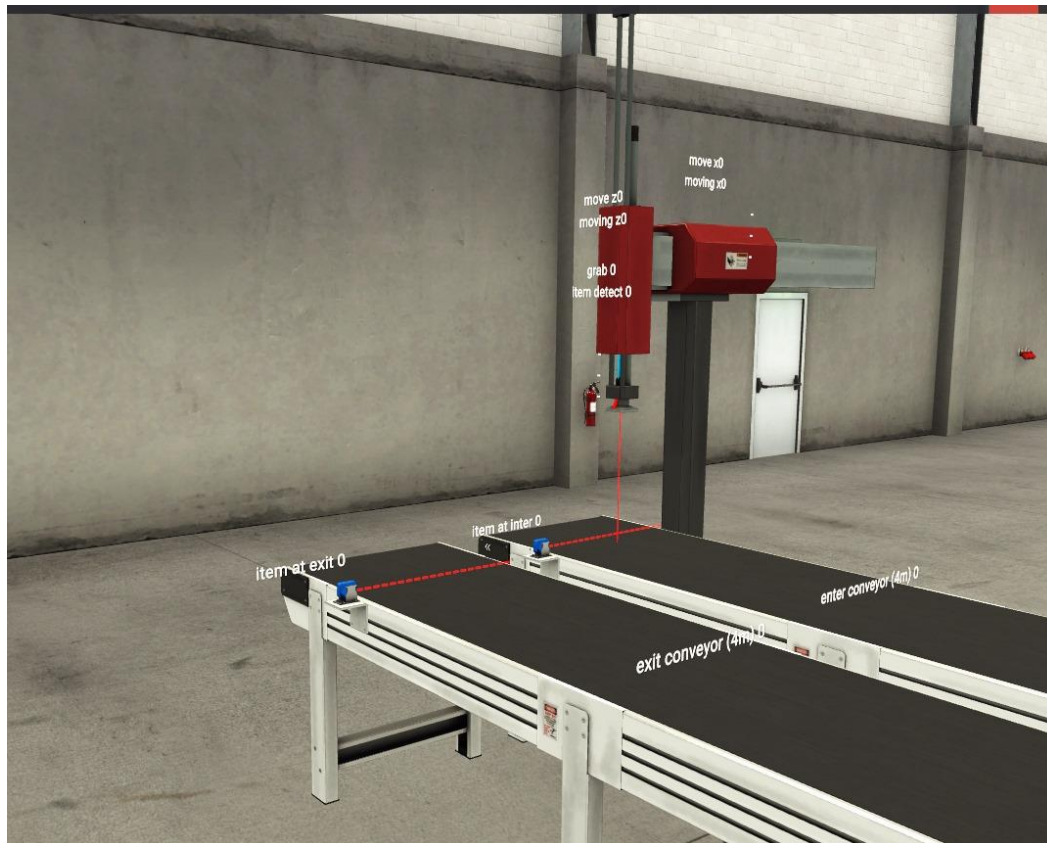


Figure 2 Feeding System

2- Sorting station 1

- for sort the green Raw Materials and Blue Raw Materials So the Green Raw Material go to CNC Machine and the blue Raw material go to another.



Figure 3 Sorting Station 1

3- Machining Center station

- The Machining center is a station used to manufacture lids and bases from raw materials. First, the articulated robot waits for raw material to be placed at the entry bay. When new material is detected, it is loaded into the CNC machine which will start manufacturing an item. Each item type takes a different interval of time to be produced (lids: 6 seconds; bases: 3 seconds). Once the operation is complete, the robot places the item on the exit bay.
- We will Use Two Machining Center Each Machine will produce Lid and Base then the next material will be Base and so on .



Figure 4 Machining Center station

4-Sorting Station 2

- This station Use for Sorting the Lids and Base .
- We Use Two of this station (one for Green and another for Blue).

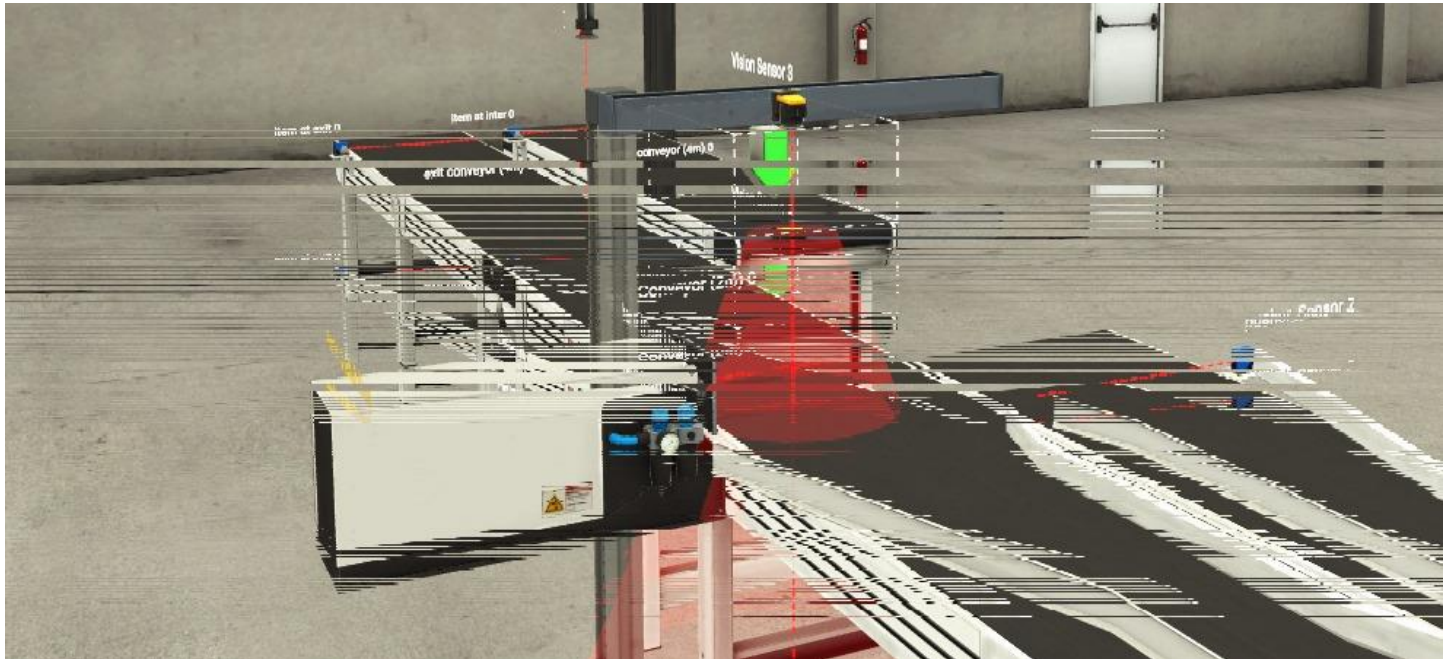


Figure 5 Sorting Station 2

5-Control Panels



Figure 6 3 Control Panels For 2 Assembler and the Production Line

6- Assembly Station

- This Station used to assemble Lids on Bases or pick and place items from one place to another.
- To guarantee a correct fit, bases and lids should be properly aligned by positioning bars.
- We Use Two of this station (one for Green and another for Blue).

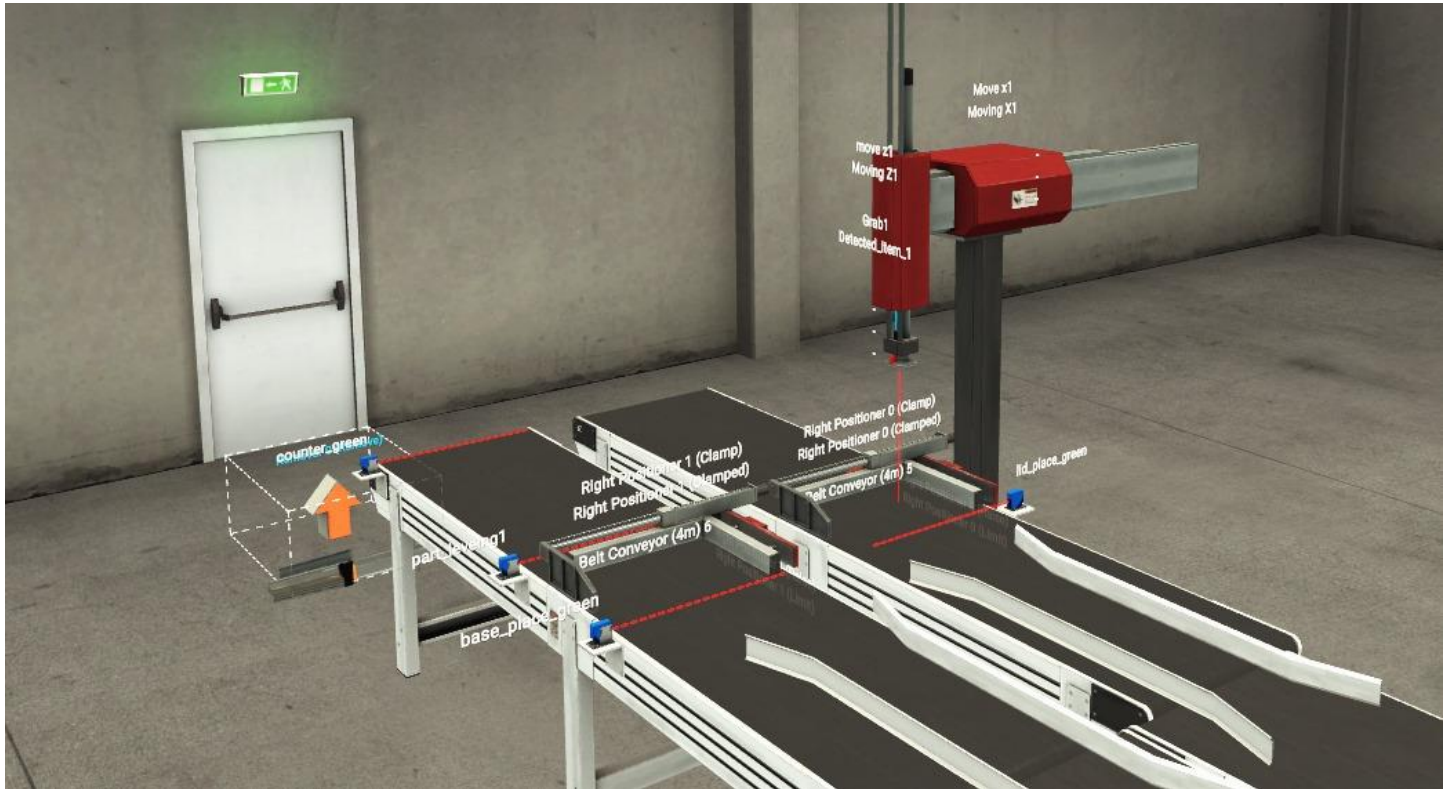


Figure 7 Assembly Station

HMI (Human Machine Interference):

We have 3 screens For :-

- 1- Green CNC
- 2- Blue CNC
- 3- Home

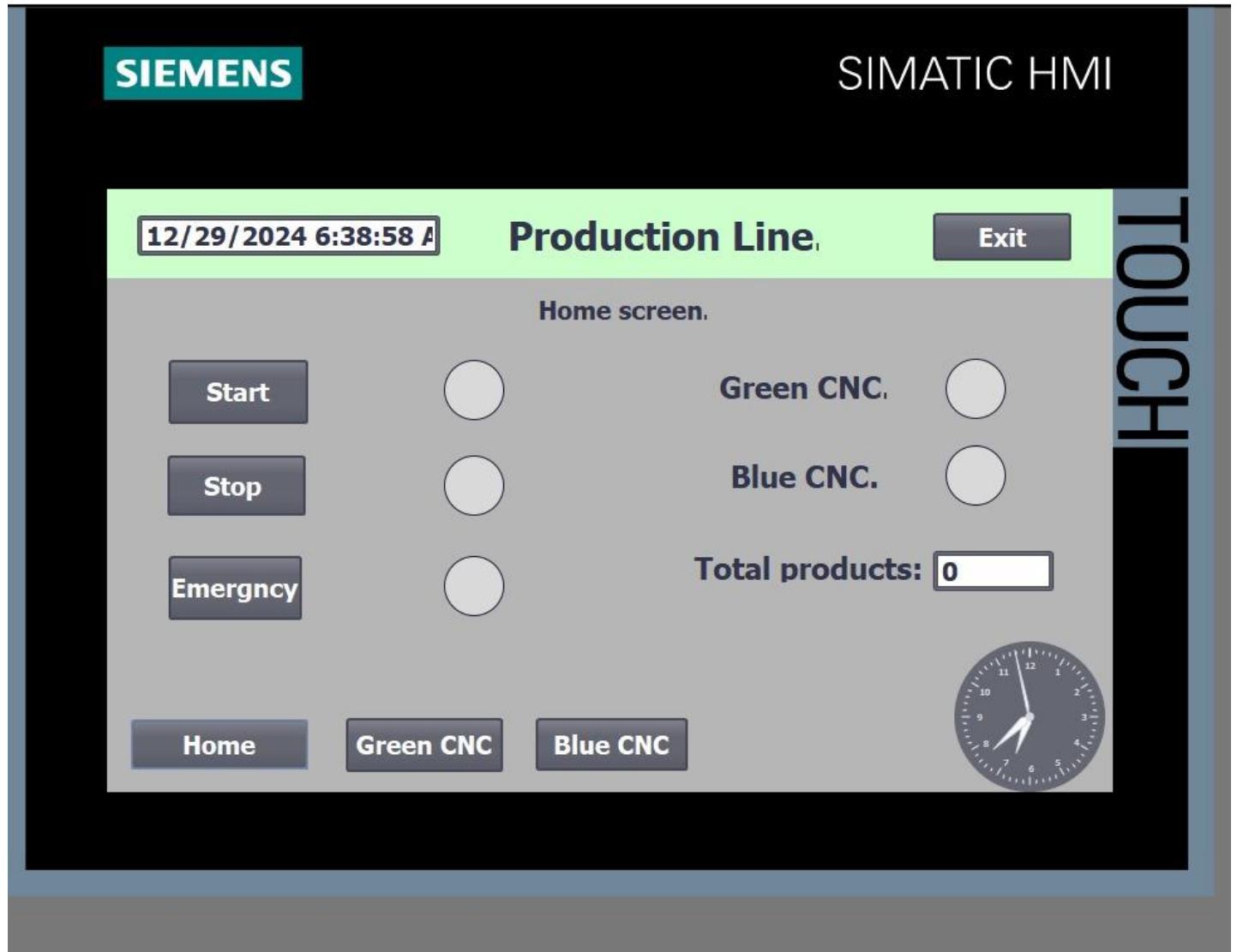


Figure 8 HMI Screen

Conclusion :-

In this Project we learnt a lot of things about the meaning of Controlling a lot of things with logic sequence .

We learnt how to design a Technical Ladder diagram that Simulated with 3 different tools .

We had some problems with simulation by PLCSIM and Factory IO that Consume a lot of time from us to solve.

We did an Extra HMI to See all States in our project:-

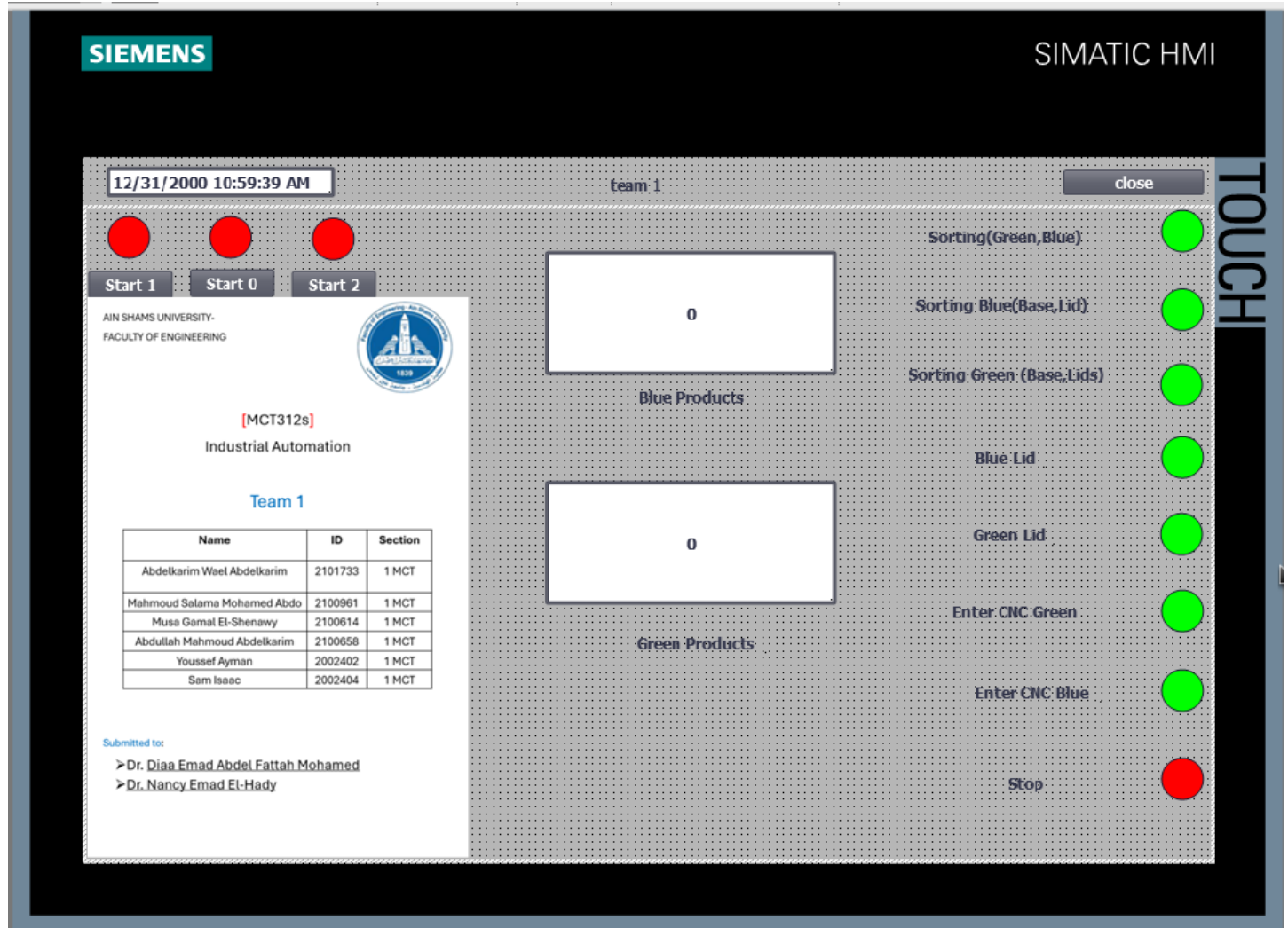


Figure 9 States HMI

Drive Links:-

We put all Ladder diagrams in PDF files and Files of the Project that simulated in front of you.

<https://drive.google.com/file/d/1SGahfPouiLPoJYkXrL-QQnOxOS9bBCqT/view?usp=sharing>